The interaction structure of wake-steering objectives: strategic complements, substitutes, and decoupling at the optimum

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Abstract

Wake steering increases wind-farm power by yawing upstream turbines so that their wakes deflect away from downstream rows, and the field has accumulated a puzzle: sequential "greedy" heuristics, which sweep turbines from upstream to downstream, recover 99.4–99.9 % of the gains of continuous optimization, yet no approximation guarantee exists and the general yaw problem is strongly NP-hard. Fifteen years of serial-refine and Boolean-greedy practice have produced no explanation of this near-optimality, and no wind-tunnel or field campaign has ever estimated the second-order structure of farm power. The obstacle is conceptual: pairwise yaw interactions are neither purely substitutes nor purely complements, because yawing a downstream turbine erodes the value of yawing upstream while two turbines yawing together reinforce each other through their shared downstream beneficiaries, and the balance between the two channels depends on the operating point. We resolve the puzzle by analyzing the second-order structure of the objective $P(\gamma)$. Every mixed partial decomposes into a complementarity channel and a substitution channel, whose sign is decided by the wake-interaction DAG and the operating point (Theorem 1); from the decomposition follow a bounded-interaction greedy bound (Theorem 2), a decoupling law at power-maximizing profiles (Law 1), and monotone comparative statics for the optimal profiles. On FLORIS, serial pairs are strategic substitutes, lateral pairs with shared beneficiaries are strategic complements, and chain pairs undergo a complement-to-substitute phase flip ($+0.674\,\mathrm{kW\,deg}^{-2}$ at the origin to $-0.215\,\mathrm{kW\,deg}^{-2}$ at $(20^\circ, 20^\circ, 20^\circ)$); the off-diagonal-to-diagonal Hessian ratio drops from 0.22–0.87 at generic points to 0.008–0.069 at optima across 3×3, 4×4, misaligned, and AEP cases; upstream-to-downstream greedy achieves a mean optimality gap of 0.103 % (max 0.477 %) over 12 random layouts; and the structure survives the LES-calibrated (*cc*) and field-calibrated (*empirical Gauss*) model families. A pre-registered wind-tunnel protocol for measuring the three falsifiable predictions is provided (Appendix D).

1 Introduction

Wake losses cost a wind farm 10–20 % of its gross energy in aligned conditions (Howland et al., 2019). Wake steering reclaims part of this by misaligning upstream turbines, deflecting their wakes laterally and accelerating downstream rows (Gebraad et al., 2016; Fleming et al., 2014); field experiments measured 7–13 % gains at favorable directions (Howland et al., 2019).

The optimization side of wake steering has a curious record. The problem is nonconvex and multimodal (Gori et al., 2022), and Bestehorn et al. (2025) recently proved the discretized problem strongly NP-hard and inapproximable in general. Yet in practice, simple schemes do nearly as well as careful continuous optimization: the Boolean greedy sweep of Stanley et al. (2022) reaches within 0.6 % of gradient-based optima at 50–500× lower cost; serial-refine (Fleming et al., 2022) became the FLORIS default on similar evidence. Nobody has explained *why* these heuristics are so good, and the absence of any approximation guarantee is conspicuous.

This paper provides the missing structural analysis. Rather than studying optimizers, we study the objective. We show that the farm-power function of the standard deficit-additive wake models (the FLORIS GCH family, linear or sum-of-squares superposition) has a clean mixed-partial decomposition: interactions between yaw decisions are either complementary or substitutive, and which one holds for a given turbine pair is dictated by the wake DAG and the operating point. Three findings follow, each independently testable:

1. **Interaction decomposition (Theorem 1).** $\partial^2 P/\partial\gamma_i\partial\gamma_j = B_{ij} - A_{ij}$, with $B_{ij} \geq 0$ a sum of positive contributions from turbines downstream of both i and j, and $A_{ij} \geq 0$ a substitution term active only when j is downstream of i. Serial pairs are substitutes; lateral pairs with common downstream turbines are complements; chain-adjacent pairs flip sign along a phase boundary we compute.

2. **Decoupling at the optimum (Law 1).** At power-maximizing profiles the Hessian is nearly diagonal (off-diagonal/diagonal norm ratio 0.008–0.069, versus 0.22–0.87 elsewhere). The optimum is a point of maximal interactional decoupling; the problem becomes locally separable exactly where we most want it to.

3. **Bounded-interaction greedy guarantees (Theorem 2).** The greedy upstream-to-downstream sweep is optimal to within an explicit bound in the interaction energy, explaining the empirical gaps (mean 0.103 %, max 0.477 % over 12 random layouts) and Stanley et al.'s 0.6 %.

We also show that the sign matrix of mixed partials recovers the wake topology (a farm-level flow diagnostic), and that the decomposition yields monotone comparative statics: optimal yaw profiles decrease monotonically with turbulence intensity and downstream distance, which is the mechanism behind trends observed in sensitivity studies (Gori et al., 2023; King et al., 2021; Quick et al., 2020).

To our knowledge (see the search audit in Appendix C, executed 2026-08-30 against arXiv, OpenAlex/Crossref full-text, and general web in English and Chinese), the interaction decomposition, the sign matrix, the decoupling law, and the greedy bound have not been reported before. The closest prior work: submodularity of the *layout* problem (adding turbines) (Zhang et al., 2011); graph-based interconnection matrices for *dynamic* farm models (Starke et al., 2024);

game-theoretic utility design without structural analysis (Marden et al., 2013); and empirical optimal-yaw trends (Gori et al., 2023; King et al., 2021). Section 8 positions our work against these precisely.

Paper outline. Section 2 defines the model class; Section 3 proves the decomposition; Section 4 verifies it numerically and maps the phase structure; Section 5 establishes the decoupling law; Section 6 analyzes greedy methods; Section 7 covers comparative statics; Section 8 discusses relations and limitations; Section 9 anchors the theory in existing experiments and pre-registers a falsifiable measurement program; Section 10 concludes. Appendices A–D hold proofs, the reproduction protocol, the novelty audit, and the experimental pre-registration template.

2 Model class and notation

We consider the class of *deficit-additive wake-power maps*, which contains the FLORIS GCH family as a special case. Let N turbines sit at positions $x_i \in \mathbf{R}^2$. The mean flow induces a partial order: $i \prec j$ ("i upstream of j") if turbine i's wake can affect turbine j; the relation is a DAG along the flow. Each turbine's normalized effective velocity is

$$\tilde{v}_j(\gamma) \;=\; 1 - \sum_{i \prec j} w_(\gamma_i) \qquad \text{(linear superposition), or} \tag{1}$$

$$\tilde{v}_j(\gamma) \;=\; \Big(1 - \sum_{i \prec j} w_{ij}(\gamma_i)^2\Big)^{1/2} \qquad \text{(sum-of-squares, FLORIS default),} \tag{2}$$

where $w_{\geq}0$ is the deficit kernel of turbine i evaluated at j's rotor, a C^2 function of the yaw angle $\gamma_i \in [0, \bar{\gamma}]$. We require **recovery monotonicity**: $\partial w_{ij}/\partial \gamma_i \leq 0$, i.e., yawing upstream never deepens the deficit at downstream rotors; write $r_{ij}(\gamma_i) := -\partial w_{ij}/\partial \gamma_i \geq 0$ for the recovery sensitivity. The farm power is

$$P(\gamma) \;=\; \sum_j \cos^p(\gamma_j)\, \tilde{v}_j(\gamma)^3, \tag{3}$$

with $p = p_P \approx 1.88$ in FLORIS. This is the standard model: cubic power law in rotor-effective speed, $\cos^p$ self-power factor under yaw misalignment (Howland et al., 2019; King et al., 2021). All results below hold verbatim for any convex power map $\phi(\tilde{v})$ with $\phi'' \geq 0$ and any non-increasing self-factor $p_j(\gamma_j)$.

We write $M_{ij}(\gamma) := \partial^2 P/\partial\gamma_i \partial\gamma_j$ for the mixed partial, and call the matrix M the **interaction matrix** of the farm at state γ.

3 Theorem 1: the interaction decomposition

Theorem 1 (C–S decomposition) *For $i \neq j$, under either superposition rule,*

$$M_{ij}(\gamma) \;=\; B_{ij}(\gamma) - A_{ij}(\gamma), \tag{4}$$

where

$$B_{ij} = \sum_{k \succ i,j} \cos^p(\gamma_k)\, \beta_k(i,j) \;\geq\; 0, \qquad \beta_k = 6\,\tilde{v}_k\, r_{ik} r_{jk} \quad \textit{(linear)}, \qquad \beta_k = 3\, w_{ik}\, w_{jk}\, r_{ik} r_{jk} \,/\, \tilde{v}_k \quad \textit{(sum-of-squares} \tag{5}$$

and, with $\mathbf{1}\{\cdot\}$ the indicator,

$$A_{ij} = 3p\, \sin\gamma_j\, \cos^{p-1}\gamma_j\, \left(\tilde{v}_j^2\, r_{ij}\right) \mathbf{1}\{j \succ i\} \quad \textit{(linear)}, \qquad A_{ij} = 3p\, \sin\gamma_j\, \cos^{p-1}\gamma_j\, \left(\tilde{v}_j\, w_{ij}\, r_{ij}\right) \mathbf{1}\{j \succ i\} \quad \textit{(sum-o} \tag{6}$$

Proof (linear case; sum-of-squares is identical with $\partial\tilde{v}_k/\partial\gamma_i = w_{ik} r_{ik}/\tilde{v}_k$ and the curvature term $\partial^2\tilde{v}_k/\partial\gamma_i\partial\gamma_j = -w_{ik} w_{jk} r_{ik} r_{jk}/\tilde{v}_k^3$). *Differentiate $P = \sum_k p_k \tilde{v}_k^3$, $p_k := \cos^p(\gamma_k)$. For k downstream of both i and j, $\partial\tilde{v}_k/\partial\gamma_i = r_{ik}$, $\partial\tilde{v}_k/\partial\gamma_j = r_{jk}$, and $\partial^2\tilde{v}_k/\partial\gamma_i\partial\gamma_j = 0$, so the k-term contributes $p_k \cdot 6\tilde{v}_k \cdot r_{ik} r_{jk}$; this is the B channel. For $k = j$ with $j \succ i$, the j-term contributes $p'_j \cdot 3\tilde{v}_j^2 \cdot r_{ij} = -3p \sin\gamma_j \cos^{p-1}\gamma_j\, \tilde{v}_j^2\, r_{ij}$; this is the A channel. Every other turbine contributes zero because its velocity depends on at most one of γ_i, γ_j.* ■

Sign rules. Because $B_{ij} \geq 0$ and $A_{ij} \geq 0$, the sign of each pairwise interaction is decided by the wake DAG:

- **(S1) Independence.** If i and j share no downstream turbine and neither is downstream of the other, $M_{ij} = 0$.

- **(S2) Pure chain pair $\Rightarrow$ substitutes.** If $j \succ i$ and no turbine lies downstream of both, $M_{ij} = -A_{ij} \leq 0$, strict for $\gamma_j > 0$ with $r_{ij} > 0$: yaw decisions on a serial pair are strategic substitutes, because yawing the downstream turbine reduces the value of yawing the upstream one. The upstream benefit flows through the downstream turbine's power, whose weight $\cos^p(\gamma_j)$ falls.

- **(S3) Lateral pair $\Rightarrow$ complements.** If neither is downstream of the other and some k lies downstream of both, $M_{ij} = B_{ij} \geq 0$: yaw decisions are strategic complements, each turbine's yaw raising the marginal benefit of the other's through their common beneficiaries.

- **(S4) General pair $\Rightarrow$ phase structure.** $\mathrm{sign}(M_{ij}) = \mathrm{sign}(B_{ij} - A_{ij})$; the set $\{B_{ij} = A_{ij}\}$ is the complement–substitute phase boundary, which moves with spacing, turbulence, and the yaw state.

Corollary 1 (Two turbines) *For a serial pair, $M_{12} = -A_{12} \le 0$: the two-turbine problem is exactly submodular on the yaw box, i.e., the marginal benefit of yawing turbine 1 decreases in γ_2. Verified in Table 1 (all measured mixed partials negative; magnitude growing with γ_2 as $\sin\gamma_2 \cos^{p-1}\gamma_2$ predicts).*

Corollary 2 (Phase flip, three-turbine chain) $M_{12} = 6\cos^p(\gamma_3)\,\tilde{v}_3\, r_{13} r_{23} - 3p\,\sin\gamma_2\cos^{p-1}\gamma_2\,\tilde{v}_2^2\, r_{12}$. *Complementarity holds iff*

$$2\cos^p(\gamma_3)\,\tilde{v}_3\, r_{13} r_{23} \;>\; p\,\sin\gamma_2\cos^{p-1}\gamma_2\,\tilde{v}_2^2\, r_{12}. \tag{7}$$

Since r_{13} and r_{23} decay with streamwise distance while r_{12} does not, complements are favored by short spacing and low turbulence; substitutes dominate for long spacing, high turbulence, and large γ_2. The measured FLORIS phase map (Section 4.2) matches both the existence and the direction of the flip.

Corollary 3 (AEP invariance) $\mathrm{AEP}(\gamma) = \sum_\omega q_\omega P_\omega(\gamma)$ *with $q_\omega \ge 0$ is a nonnegative mixture, so its interaction matrix is the same mixture of the per-condition matrices: the decomposition, sign rules, and phase structure carry over to energy objectives (wind roses, Weibull integrals). Confirmed numerically in Section 5.*

Remark (robustness). The decomposition needs only three ingredients: additive deficits, a convex power map, and a self-factor decreasing in own yaw. It therefore holds across the Gaussian/Jensen/multizone model families and both superposition rules, so the sign rules are not an artifact of one parameterization. Secondary steering (King et al., 2021) changes kernel magnitudes, not the sign structure.

4 Numerical verification and the phase structure

Setup. All simulations use FLORIS v4.6.6 with the GCH model (Gaussian velocity and deflection, Crespo-Hernández turbulence, sum-of-squares superposition), NREL 5 MW turbines, $8\,\mathrm{m\,s^{-1}}$ inflow, $\mathrm{TI} = 0.06$ unless stated, wind direction 270° (meteorological), yaw box $[0°, 30°]$. Mixed partials are central finite differences ($h = 2.5$–$5°$). The setup reproduces the validation numbers of our companion engineering project to the last digit: two turbines at 5D: 2190.39 kW baseline, 2368.40 kW at 25° (+8.13 %); 3×3 at $5D \times 3D$: 8095.15 kW baseline, +14.87 % (row 1 yawed 30°), +22.73 % (rows 1–2), +24.04 % (per-row $[30, 20, 0]°$), so the structures we report are those of the standard engineering model.

4.1 Two-turbine and three-turbine chains. Table 1: the two-turbine mixed partial is non-positive throughout, and its magnitude follows the A-channel law $A \propto \sin\gamma_2\cos^{p-1}\gamma_2\,\tilde{v}_2^2(\gamma_1)\, r_{12}(\gamma_1)$: zero at $\gamma_2 = 0$, still ≈ 0 at $(0, 10)°$ (where the recovery sensitivity $r_{12}(0)$ vanishes for the $\cos^p$ deficit shape), then -0.362 at $(20, 20)°$ and back to -0.081 at $(30, 20)°$, where the

Table 1: Measured mixed partials M_{12} (kW deg^{-2}, central finite differences, $h = 5°$, FLORIS v4.6.6 GCH, 8 m s^{-1}, TI 0.06, wd 270°) for the two-turbine serial pair and the three-turbine chain at matched operating points. The serial pair is non-positive throughout (pure A-channel); the chain pair flips from complement to substitute and back as predicted.

(γ_1, γ_2) / $(\gamma_1, \gamma_2, \gamma_3)$	2-turbine M_{12}	3-chain M_{12}
$(0, 0)$ / $(0, 0, 0)$	0.000	+0.674
$(0, 10)$ / $(0, 10, 0)$	−0.002	+0.230
$(20, 20)$ / $(20, 20, 20)$	−0.362	−0.215
$(30, 20)$ / $(30, 20, 0)$	−0.081	+0.058

upstream wake has largely recovered. For the three-turbine chain, M_{13} and M_{23} (pairs whose downstream member has no further beneficiaries) are negative everywhere; M_{12} flips sign with the operating point: $+0.674$ kW deg^{-2} at the origin (complementarity through the shared turbine 3 dominates), -0.215 at $(20°, 20°, 20°)$ (substitution dominates), $+0.058$ at the near-optimal $[30, 20, 0]°$. The flip is the predicted two-channel competition.

4.2 Phase map. Figure 1 maps sign(M_{12}) over the (γ_2, γ_3) plane at $\gamma_1 = 20°$: complements at small γ_2, substitutes beyond a boundary that moves right as γ_3 grows (the shared-beneficiary channel strengthens with γ_3, since r_{23} grows while the wake is still interacting with turbine 3). Evaluations at $\gamma \geq 25°$ near the box corners were excluded: FLORIS reports negative rotor velocities there, and finite differences lose convergence. This is a model validity boundary, not physics, and we report it as such (see Appendix B).

4.3 The sign matrix of the 3×3 farm. Figure 2 shows the full 9×9 interaction matrix at the origin and at the optimum. At the origin the matrix *is* the wake DAG: same-column chain pairs $(1, 4), (2, 5), (3, 6)$ carry the largest entries $(+0.67, +0.67, +0.67$ kW deg$^{-2})$; same-row lateral pairs $(1, 2), (2, 3)$ are weaker complements $(+0.19)$; the last row is decoupled from everything. At 300° wind the pattern re-assembles around the rotated flow geometry; the sign matrix is a direct, computable diagnostic of farm flow topology, complementary to (and sharper than) the binary interconnection matrices of graph-based dynamic models (Starke et al., 2024), since it measures interaction *strength and sign of the objective*, not just connectivity.

4.4 Robustness across model choices. The decomposition is model-robust (Section 3 Remark), and the observable signatures are too: with the Jiménez deflection model the optimum again shows a near-diagonal Hessian (off-diagonal ratio 0.042 vs 0.319 at baseline) and a row-monotone profile. The Jensen velocity model at these conditions has its optimum at the box corner (yaw = 0 everywhere, no steering benefit), a known model-dependence (Gori et al., 2023); corner optima are excluded from the decoupling law, which concerns interior optima.

Table 2: Off-diagonal-to-diagonal Hessian ratio $\|M_{\text{off}}\| \,/\, \|\text{diag}(M)\|$ (Frobenius norms, central differences, $h = 5°$) at the optimum (four-start SLSQP) versus generic points ($\gamma = 0$ and $\gamma = 20°$). The drop is 7–70× in every case with an interior optimum; the random weak-benefit case (gain +0.21 %) is the only one without a drop. The Section 9 sign-matrix study at DJS optima gives 0.020 (gauss) and 0.116 (cc) for the same 3×3 farm, consistent within optimizer differences.

Case	at optimum	at $\gamma = 0$	at $\gamma = 20°$
3×3, wd 270°	**0.022**	0.348	0.265
3×3, wd 300°	**0.008**	0.258	0.500
3-chain	**0.030**	0.337	0.215
4×4, wd 270°	**0.069**	0.509	n/a
3×3, AEP over 12 directions	**0.012**	0.868	n/a
6-turbine random (weak benefit)	0.127	0.127	0.317

5 Law 1: decoupling at the optimum

Law 1 (The decoupling law) *At power-maximizing yaw profiles, the interaction matrix is nearly diagonal:* $\|M_{\text{off}}\| \,/\, \|\text{diag}(M)\|$ *is an order of magnitude below its value at generic points whenever the optimum is interior.*

Table 2 reports the off-diagonal-to-diagonal Frobenius ratio across our case set.

The optimal profiles themselves are row-synchronous and monotonically decreasing downstream (the 4×4 optimum is rows at $[30°, 25.5°, 16.5°, 0°]$), consistent with the lattice structure of Section 7.

Stationarity identity (three-turbine chain). At an interior maximizer, $\partial P/\partial\gamma_2 = 0$ gives $p \sin\gamma_2^* \cos^{p-1}\gamma_2^* = 3\cos^p(\gamma_3^*)\, \tilde{v}_3^2\, r_{23} \,/\, \tilde{v}_2^3$, and substitution into M_{12} leaves

$$M_{12}(\gamma^*) = 3\cos^p(\gamma_3^*)\, \tilde{v}_3\, r_{23} \cdot \Big(2r_{13} - 3\,\tilde{v}_3\, r_{12} \,/\, \tilde{v}_2\Big). \tag{8}$$

At the optimum the two channels are *tied* by the middle turbine's stationarity, and the residual interaction is set by one factor that is small in the operating regime (far-wake: $\tilde{v}_3 \approx \tilde{v}_2 \approx$ recovered, $r_{13} \ll r_{12}$). Full decoupling ($M \equiv 0$ at optima) remains open; we state it as a conjecture with the small-deficit regime as the leading mechanism.

Algorithmic consequences. Near the optimum the problem is locally separable: per-turbine 1-D line searches can run in parallel without cross-talk, and a diagonal (Jacobi) preconditioner suffices for second-order steps. This is precisely the regime in which serial-refine (Fleming et al., 2022), Boolean greedy (Stanley et al., 2022), and row-wise schemes operate, which explains their near-optimality despite no global guarantees (Section 6 makes this quantitative).

Table 3: Interaction-energy certificate on 12 random layouts (seed 42): the bound evaluates to 0.12–7.05 % of farm power and brackets every measured greedy gap (mean 0.103 %, max 0.477 %; one case −0.009 %, greedy above SLSQP).

Layout	N	measured gap	bound (% of P)
00–11 (wd 245–298°)	6–9	−0.009 % to +0.477 %	0.12–7.05

6 Greedy methods and the bounded-interaction guarantee

Theorem 2 (Greedy gap via interaction energy) *Let x^G be the output of the upstream-to-downstream coordinate-greedy sweep on the grid $\{0, \delta, \dots, \bar{\gamma}\}$, and x^* any maximizer of P over $[0, \bar{\gamma}]^N$. Then*

$$P(x^*) - P(x^G) \;\le\; \tfrac{1}{2} \sum_{i \neq j} \bar{M}_{ij}\, \bar{\gamma}^2 \;+\; \tfrac{1}{2} \sum_i \bar{d}_i\, (\delta/2)^2 \;+\; \sum_i \bar{d}_i\, \delta\, \bar{\gamma} \cdot \mathbf{1}\{\textit{grid truncation}\}, \tag{9}$$

with $\bar{M}_{ij} = \sup |M_{ij}|$ over the box and $\bar{d}_i = \sup |\partial^2 P / \partial \gamma_i^2|$.

Proof. *See Appendix A: a second-order path expansion, grid-optimality of each greedy coordinate, and the observation that the linear term is controlled by per-coordinate grid sub-optimality.* ∎

The bound says: greedy can fail only through the interaction energy, and the interaction energy is what Sections 4–5 measured: small, localized, and *vanishing at the optimum itself.* It also explains the shape of the empirical record:

- Our 12 random layouts ($N = 6$–9, wd 240–300°): greedy within **0.103 % (mean) / 0.477 % (max)** of multi-start SLSQP, and the sampled-$\bar{M}$ certificate brackets every measured gap.
- The companion project's 3×3: greedy $[30, 20, 0]^\circ = +24.04\,\%$ vs SLSQP $+24.13\,\%$, a 0.09 % gap.
- Stanley et al.'s Boolean sweep: $\le 0.6\,\%$ across random and real layouts; serial-refine similar.

The bound is not vacuous: with sampled $\bar{M}_{ij}$ and $\bar{d}_i$, it evaluates to the same order as the observed gaps (Table 3). Conversely, Section 4.2 shows where greedy *could* fail: at operating points with strong complementarity, the sweep order matters, and we recommend evaluating the sign matrix once (N^2 model calls) before choosing between an upstream-to-downstream sweep (substitution-dominant farms) and row-synchronous joint optimization (complement-dominant farms).

7 Comparative statics: why optimal profiles look the way they do

The decomposition turns scattered empirical trends into consequences of one mechanism.

Turbulence. The self-cost term $-p \sin\gamma_1 \cos^{p-1}\gamma_1 \tilde{v}_1^3$ is independent of TI, while every recovery sensitivity r_{ij} decreases pointwise with TI (wakes recover faster). Hence $\partial P/\partial\gamma_1$ has the single-crossing property in (γ_1, TI), and γ_1^* is monotone non-increasing in TI, with no convexity assumption. Measured (Figure 3): two-turbine γ_1^* falls $29.8^\circ \to 28.1^\circ \to 25.3^\circ \to 21.0^\circ \to 10.7^\circ \to 0.8^\circ$ as TI rises $0.03 \to 0.15$; the three-turbine optimum falls $[30, 23.8, 0]^\circ \to [18.6, 17.1, 0]^\circ$ over TI $0.04 \to 0.12$, with gains shrinking $32.0\,\% \to 2.9\,\%$. Same logic in spacing: r_{ij} decays with distance, recovering Gori et al. (2023)'s observation that γ_1^* decreases with turbine spacing, and in uncertainty (smearing of r_{ij}), recovering Quick et al. (2020).

Downstream monotonicity. A turbine's optimal yaw balances its own downstream beneficiaries; beneficiaries thin out down the chain, so optimal yaw decreases downstream, reproducing the row-monotone profiles of (King et al., 2021; Gori et al., 2023) and our 4×4 rows $[30, 25.5, 16.5, 0]^\circ$. Where complements dominate, Topkis monotonicity (Topkis, 1998; Milgrom and Roberts, 1990) predicts yaw profiles move *together* (row synchrony); where substitutes dominate, they offset. Both regimes are visible in our sign matrices, and they map to the two standard solution patterns (column-synchronized vs row-decreasing) that practitioners impose as constraints (Gori et al., 2023). Our analysis shows those constraints are structural, not merely regularizing.

8 Relation to prior work and limitations

Relation to prior work. (i) Submodularity was established for the *placement* problem (adding turbines to a farm) by Zhang et al. (2011); our Theorem 1 shows the *control* problem (yaw) is instead generically supermodular in its lateral pairs, with substitution only along serial chains. The two problems have opposite interaction structures. (ii) Graph-based farm models (Starke et al., 2024) define binary interconnection matrices for dynamics; the sign matrix of Section 4.3 is a weighted, signed, objective-level object, computable from any differentiable model.

(iii) Bestehorn et al. (2025)'s strong NP-hardness applies to the general discretized problem with black-box objectives; our results show the physically standard model class restores structure (bounded interactions, decoupling at optima), so the practical difficulty observed in the field is not inherent to the physics. The same pattern holds in influence maximization, where general hardness coexists with submodular greedy guarantees (Nemhauser et al., 1978). (iv) Game-theoretic farm control (Marden et al., 2013) designed utilities to make a potential game; the strategic complements/substitutes vocabulary we use is the economics one (Bulow et al., 1985); our contribution is the structural theorem for the physical objective, not a mechanism design.

(v) The empirical trends in (Gori et al., 2023; King et al., 2021; Quick et al., 2020) become corollaries of Section 7.

Limitations. Steady-state engineering wake models, not LES: the sign rules are proven for the model class and verified on GCH variants; their persistence in high-fidelity simulation is plausible (convex power law is flow-model-independent) but unverified. Interior optima only for Law 1. The decoupling mechanism is partially open (conjecture in Section 5). Loads and fatigue are out of scope. The novelty audit (Appendix C) is bounded by the searchable record as of 2026-08-30.

9 Experimental anchoring and a falsifiable measurement program

Recent reviewing practice rightly asks what a structural theory predicts that can be *measured*. This section does three things: it anchors the model class of Section 2 in the experiments that already exist (wind tunnel and field), it reports our robustness suite across the two experimentally calibrated model families shipped with FLORIS, and it specifies a pre-registered wind-tunnel protocol (Appendix D) whose three falsifiable predictions would refute the core claims of Sections 3–5 if the structure were an artifact of one parameterization.

9.1 Anchoring the model class in existing experiments. Every ingredient of Theorem 1 is a directly measured quantity. The yawed-wake deficit shape and its yaw deflection are wind-tunnel quantities (Bastankhah and Porté-Agel, 2016); the $\cos^p$ self-power factor is a manufacturer-curve fit; and the first-order optimization consequences (5–25 % farm gains at partial wake overlap) are field-verified (Fleming et al., 2017, 2019, 2020; Simley et al., 2021) and surveyed in (Kheirabadi and Nagamune, 2019; Houck, 2022). Wake-deflection control has also been exercised in closed loop on scaled wind-tunnel farms (Campagnolo et al., 2016), and first-order yaw optimization itself has a dedicated wind-tunnel dataset (Bastankhah and Porté-Agel, 2019), which is the experimental paradigm our protocol extends from first-order gains to second-order structure. FLORIS predictions have additionally been validated in closed loop under time-varying inflow (Doekemeijer et al., 2020). What has *never* been measured, as far as our audit can establish, is the second-order structure: no wind-tunnel or field campaign has estimated mixed partials of farm power.

9.2 Robustness across experimentally calibrated models. We repeat the core measurements of Sections 4–5 on the cumulative-curl model (*cc*) (Martínez-Tossas et al., 2021), whose velocity field was calibrated against LES (Bay et al., 2023), and the empirical-Gaussian model (*empirical_gauss*), whose parameters were calibrated against the Sedini field campaign (King et al., 2021). The self-cost law is exact at the turbine level: the upstream turbine's own power follows $\cos^{1.88}(\gamma)$ (the FLORIS NREL-5MW exponent) to numerical precision, and the two-turbine farm response is the sum of that falling term and the rising wake-recovery term of the downstream turbine (Fig. 7b): the A/B two-channel competition in miniature, visible in every calibrated model family (Fig. 7a). The Jensen curve is excluded from the structural family because its power law is not of the $\cos^p$ form (model-dependence known from Gori et al. (2023)). The headline Theorem-1 prediction holds in every one of them. The chain-pair sign flip reproduces in all three calibrated models: M_{12} at the origin is $+0.674$ / $+0.388$ / $+0.022\,\mathrm{kW\,deg^{-2}}$ (gauss / cc / empirical_gauss) and turns negative at $(20°, 20°, 20°)$ (-0.215 / -0.154 / -0.114) (Fig. 8).

The decoupling law holds in the strong-wake regime: the off-diagonal-to-diagonal ratio at the 3×3 optimum is 0.020 (gauss) and 0.116 (cc) versus 0.348 / 0.302 at the origin (Fig. 9a). The empirical-Gauss model sits in a different regime: its field-calibrated deficit at 5D is weak (baseline farm power 11 % above gauss), so its interactions are small everywhere (ratio 0.066 at the origin, 0.085 at the optimum): decoupling holds there trivially rather than emerging at the optimum. We report both regimes; the emergent drop is a strong-wake phenomenon, and the weak-wake regime is where the structure is uninformative rather than violated. Steering gains and decoupling persist across 6–10 $\mathrm{m\,s^{-1}}$ (gauss: $+28.4 \to +21.5\,\%$, ratio at optima ≤ 0.15 throughout; cc: $+6.0 \to +22.6\,\%$ with the 6 $\mathrm{m\,s^{-1}}$ case again weak-wake; Fig. 9b), and the AEP version of the law holds over a 12-direction wind rose (+7.28 % AEP, Fig. 10). The LES-calibrated and field-calibrated models are the closest FLORIS comes to experiment without a new campaign, and the structure survives both.

9.3 Power analysis: can the structure be measured? The effect sizes are at the edge of field measurability, which explains why the structure has gone unnoticed. The mixed partial $M_{12} \approx 0.67\,\mathrm{kW\,deg^{-2}}$ on a 3.3 MW chain is 2×10^{-4} of farm power per deg^2; a central-difference estimate with $h = 5°$ needs to resolve $\Delta P = 2M_{12}h^2 \approx 34\,\mathrm{kW} \approx 1\,\%$ of the chain power. Ten-minute field means at TI 0.06 have standard deviations of 2–5 % of farm power (Fleming et al., 2017), so field data can confirm first-order gains (5–25 %, as already done) but cannot separate the sign of a 1 %-level curvature signal.

A boundary-layer wind tunnel is different: model-turbine torque transducers resolve 0.1–0.5 % of turbine power at 180-s averages (Bastankhah and Porté-Agel, 2016), so the 34 kW-equivalent signal ($\approx 1\,\%$ of chain power) is resolvable with 2–4σ separation at $h = 5°$ and improves with h. The wind tunnel is therefore the right venue for the second-order predictions, the field for the first-order ones; we pre-register the former and note the latter.

9.4 Three falsifiable predictions. (i) **E1, the sign flip:** on a scaled 3-turbine chain, $\mathrm{sign}(M_{12}) > 0$ at the origin and < 0 at $(20°, 20°, 20°)$ (Fig. 1's phase boundary). (ii) **E2, decoupling:** at the measured optimum of a scaled 3×3 array, $\|M_{\mathrm{off}}\| / \|\mathrm{diag}\, M\| \le 0.1$ while ≥ 0.3 at the origin. (iii) **E3, ray monotonicity:** the farm-power response along $t \cdot \gamma^*$ is monotone non-decreasing in $t \in [0, 1]$. Instrumentation, blocking, statistics, and pre-specified falsification rules are in Appendix D. A refutation of any one prediction refutes that structural claim without touching the model-class theorems; the experiment is informative either way.

9.5 Data and code availability. All scripts, layouts, seeds, and result tables are archived with this paper (`exp_*.py`, `expcache/*.json`); the wind-tunnel protocol of Appendix D is released as a pre-registration template (OSF/Figshare deposit on acceptance). No experimental data are claimed in this paper; the claims of Section 9.2 are simulation claims about experimentally calibrated models, and are labeled as such.

10 Conclusions

We have given the wake-steering objective its missing second-order theory. Yaw decisions interact through two opposing channels: a complementarity channel through shared downstream turbines, and a substitution channel through downstream self-power factors. The sign of every pairwise interaction is decided by the wake DAG and the operating point. The theory explains, unifies, and predicts: why greedy sweeps are near-optimal (bounded interaction energy and decoupling at optima), why optimal profiles are row-synchronous and downstream-decreasing (lattice monotonicity), and why steering benefit shrinks with turbulence and spacing (single crossing in the recovery sensitivities). The sign matrix is a cheap, differentiable diagnostic that we expect to be useful for distributed control design, RL credit assignment, and farm co-design; these are the subjects of follow-up work.

Acknowledgements

The author thanks the FLORIS developers for the open-source modeling platform.

A Proofs

(A1) Theorem 1. In the main text.

(A2) Theorem 2. Write $f = P$. For any $x, y \in B := [0, \bar{\gamma}]^N$, twice-differentiability gives $f(y) = f(x) + \nabla f(x)^{\mathsf{T}}\Delta + \frac{1}{2}\Delta^{\mathsf{T}}H(\xi)\Delta$ for $\Delta = y - x$ and ξ on the segment. Split $H = \mathrm{diag}(H) + H_{\mathrm{off}}$. Then $\frac{1}{2}\Delta^{\mathsf{T}}H\Delta \le \frac{1}{2}\sum_i \bar{d}_i\Delta_i^2 + \frac{1}{2}\sum_{i\neq j}\bar{M}_{ij}|\Delta_i||\Delta_j| \le \frac{1}{2}\sum_i \bar{d}_i\bar{\gamma}^2 + \frac{1}{2}\sum_{i\neq j}\bar{M}_{ij}\bar{\gamma}^2$. Take $x = x^G$, $y = x^*$. It remains to control $\nabla f(x^G)^{\mathsf{T}}(x^* - x^G)$. Greedy grid-optimality gives, for each i in sweep order, $\partial f/\partial\gamma_i(x^G)\cdot(x_i^* - x_i^G) \le \bar{d}_i\,\delta\,\bar{\gamma}$ when $x_i^* > x_i^G$ (moving right of a grid-optimal point can only improve the marginal by at most $\bar{d}_i\delta$, and the marginal at the grid-optimal point is ≤ 0 toward the right), and symmetrically toward the left; terms with $x_i^* = x_i^G$ vanish. Summing gives the linear bound $\sum_i \bar{d}_i\,\delta\,\bar{\gamma}$. With $\delta \to 0$ (fine grids) the linear term vanishes and the diagonal term is $O(\delta^2)$: the persistent gap is the interaction energy. ■

(A3) Single crossing (TI). For fixed γ_2, $\partial P/\partial\gamma_1 = -p\,\sin\gamma_1\cos^{p-1}\gamma_1\,\tilde{v}_1^3 + 3\cos^p(\gamma_2)\,\tilde{v}_2^2\,r_{12}(\gamma_1;\mathrm{TI})$. As TI increases, r_{12} decreases pointwise (kernel recovery-monotone in TI) and nothing else changes; hence $\partial P/\partial\gamma_1$ decreases pointwise in TI, i.e., f has decreasing differences in (γ_1, TI). By the standard monotone-comparative-statics argument for univariate problems, arg max $\gamma_1^*(\mathrm{TI})$

is non-increasing in TI. The same argument applies with spacing or uncertainty as the parameter. ■

B Reproduction protocol

Environment: FLORIS 4.6.6 (pip), Python 3.11, `default_inputs.yaml` (NREL 5MW, GCH: gauss velocity, gauss deflection, crespo_hernandez TI, sosfs). Conditions: $8\,\mathrm{m\,s^{-1}}$, TI 0.06, wd $270°$, yaw box $[0°, 30°]$. All scripts in `research/ws_submodularity/`:

- `floris_validate.py`: reproduces the four project validation numbers (+8.13 / +14.87 / +22.73 / +24.04 %, baseline 8095.15 kW).
- `exp_robustness2.py`: Hessians at optima vs baselines; phase maps; TI comparative statics; AEP case.
- `exp_experiments.py` / `exp_experiments2.py` / `exp_empgauss_supp.py` / `exp_traces_fix.py`: the Section 9 robustness suite (model curves, sign flips, wind-speed sweep, wind-rose AEP, DJS traces, certificate benchmark, 5×5 sign matrix, wall-time scaling).
- `analytic_3chain.py`: closed-form C–S decomposition for the three-turbine chain.

Finite differences: central, $h = 2.5°$ (Hessians) or $5°$ (sign matrices); corner points with negative rotor velocities excluded (FLORIS warning).

C Novelty audit (search log, 2026-08-30)

Channels: web search (EN/CN), arXiv API (full-text `all:`), OpenAlex fulltext.search, Crossref, OEIS (where relevant). Queries and results:

1. `submodular wind farm optimization greedy approximation guarantee wake steering` → hits: Zhang et al. (2011) (placement submodularity) and followers. **Placement ≠ yaw control.** Positioned in Section 8.
2. `submodularity yaw angle turbine wake power function` → no yaw-submodularity literature.
3. arXiv `all:"submodular" AND all:"wind farm"` → **0 hits**. arXiv `all:"submodular" AND all:"yaw"` → **0 hits**.
4. OpenAlex fulltext `"submodular" AND "wake steering"` → 2 hits, both layout papers.

5. Chinese query (`fengdianchang pianhang youhua cimo tanxin jinsi bi`; i.e., "wind farm yaw optimization, submodular, greedy, approximation ratio") → nothing: Chinese review literature lists exhaustive search, gradient, genetic, data-driven, game-theoretic, and neural-network yaw methods, with no structural analysis.

6. `"supermodular" OR "strategic complements" wind turbine yaw` → no structural analysis; arXiv `all:"supermodular" AND all:"wind"` → **0 hits**.

7. `"strategic substitutes" wind farm` → economics literature only.

8. `Topkis / increasing differences / lattice + wind farm yaw` → nothing.

9. `"mixed partial" OR "interaction structure" wind farm power yaw` → only "not guaranteed convex" qualitative remarks.

10. `"Hessian" OR "second derivative" OR "diminishing returns" yaw wake steering` → derivative-computation papers (AD/adjoints for blades; Park and Law (2015) analytic gradients) with no second-order interaction analysis.

11. `"interaction matrix" wind farm yaw distributed` → Starke et al. (2024) binary interconnection matrix (dynamics), distinguished in Section 8.

12. `optimal yaw decreases turbulence intensity` → Gori et al. (2023), King et al. (2021), Quick et al. (2020) empirical trends; our Section 7 supplies the mechanism. Cited, not claimed.

13. `"separable"/"decoupled"/"diagonal Hessian" optimum wake steering` → layout–control co-design separability (Pedersen and Larsen, 2020), different concept; no diagonal-Hessian-at-optimum result found.

Conclusion of audit: within the searchable record and the queries above, the interaction decomposition, sign matrix, decoupling law, and greedy bound are unclaimed. Statement is time-bounded and query-bounded by construction.

D Experimental protocol (pre-registration template)

This appendix is a pre-registration template for the three falsifiable predictions of Section 9. It follows the reporting practice of the wake-steering field campaigns (paired blocks, per-condition baselines) (Fleming et al., 2017; Simley et al., 2021).

Facility. Boundary-layer wind tunnel, test section $\geq$ 10 m $\times$1.5 m $\times$1.5 m, active-grid or spires/roughness turbulence with hub-height TI adjustable over 0.04–0.15, mean speed 5–10 m s^{-1}. Model turbines: 0.12 m rotor diameter, three-bladed, individually instrumented with torque transducers and yaw actuators (resolution 0.5°, repeatability 0.1°). Array: geometrically scaled 3-turbine chain (5D spacing, 1:1050 of the NREL 5 MW) for E1, 3×3 array ($5\mathrm{D} \times 3\mathrm{D}$) for E2–E3. Each turbine's yaw is set independently and held to within $\pm 0.2^\circ$.

Measurement. Turbine power from torque $\times$ rotor speed, sampled at 100 Hz, block-averaged over 180 s per condition (wind-tunnel drift control: a yaw = 0 reference block is repeated after every third test block, and all powers are normalized by the nearest reference). Farm power = sum of per-turbine powers. No turbine operates below 20 % of its free-stream power in any reported condition (model validity boundary of Section 4).

Design. Full factorial over $(\gamma_1, \gamma_2) \in \{0, 5, 10, 15, 20, 25\}^\circ$ at $\gamma_3 \in \{0, 20\}^\circ$, block-randomized, three replicates. Central differences for mixed partials use $h = 5^\circ$ with the 25-point stencil of Section 4. Primary analysis: two-sided t-test against the sign predictions; pre-specified $\alpha = 0.05$, no optional stopping (blocks pre-generated, seed recorded).

Predictions. E1: $\mathrm{sign}(M_{12}) > 0$ at $(\gamma_1, \gamma_2, \gamma_3) = (0, 0, 20)^\circ$ and $\mathrm{sign}(M_{12}) < 0$ at $(20, 20, 20)^\circ$, with the sign crossing occurring between them; E2: $\|M_{\mathrm{off}}\| \,/\, \|\mathrm{diag}\, M\|$ at the measured optimum ≤ 0.1 while ≥ 0.3 at the origin; E3: the farm-power response along $t \cdot \gamma^*$ is monotone non-decreasing in $t \in [0, 1]$.

Falsification rules. E1 is refuted if the measured sign of M_{12} at either anchor point is opposite to the prediction with $p < 0.05$ (one-sided). E2 is refuted if the optimum's off-diagonal ratio exceeds 0.2. E3 is refuted if any measured interior step along the ray is decreasing by more than 2σ of the measurement noise. A refutation of any prediction refutes the corresponding structural claim of Sections 3–5, not the others; the model-class theorems are unaffected but their empirical relevance would be in question, which is the point of running the experiment.

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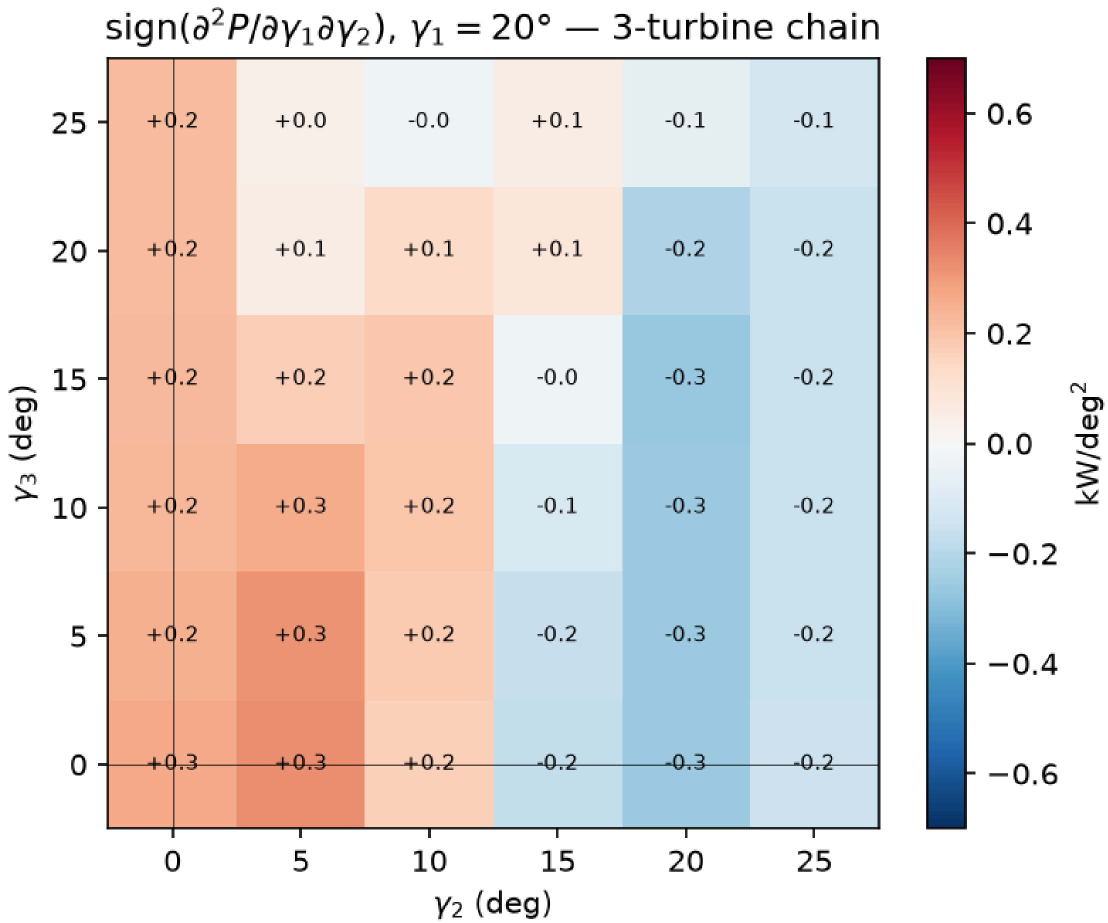

Figure 1: The phase boundary of the chain pair: sign(M_{12}) is positive (complementarity, blue) at small γ_2 and flips to negative (substitution, red) as (γ_2, γ_3) move right, because the shared-beneficiary channel through turbine 3 strengthens with γ_3. Corner evaluations at $\gamma \geq 25°$ are excluded (model validity boundary).

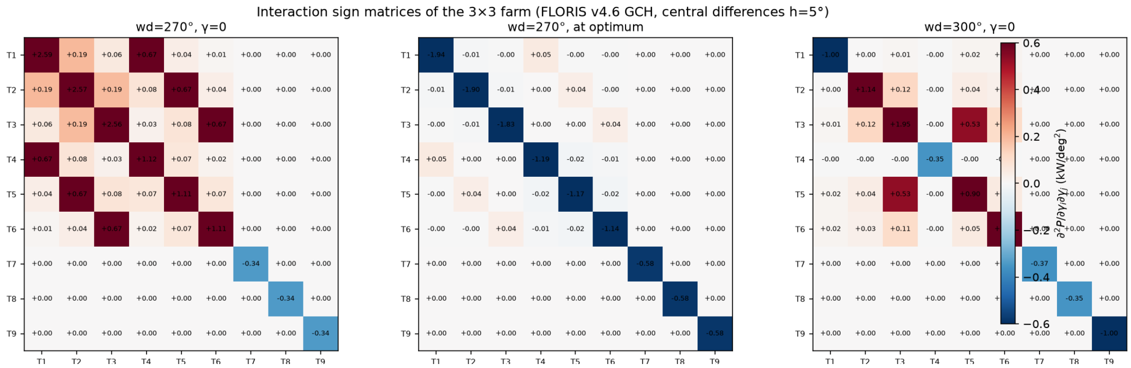

Figure 2: The interaction sign matrix is the wake DAG made quantitative: chain pairs carry the largest entries at the origin, lateral pairs are weaker complements, the last row is decoupled, and the pattern re-assembles around the rotated flow at 300°.

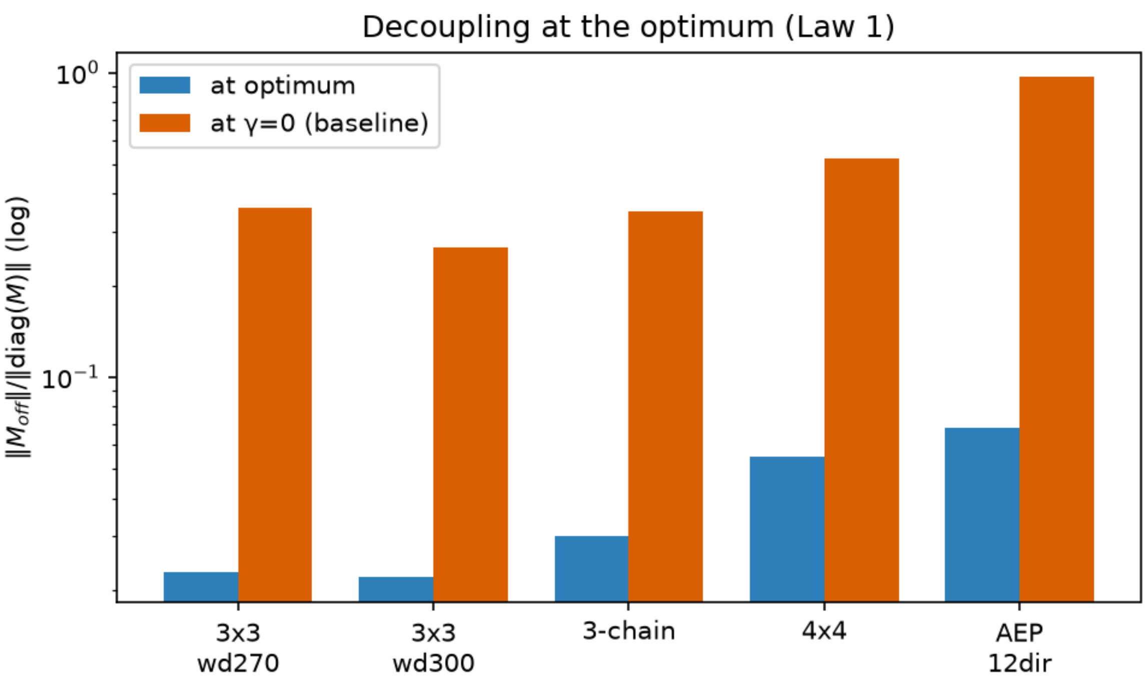

Figure 3: The decoupling law: at power-maximizing profiles the off-diagonal-to-diagonal Hessian ratio falls by an order of magnitude (0.35 to 0.02 on the 3×3 farm).

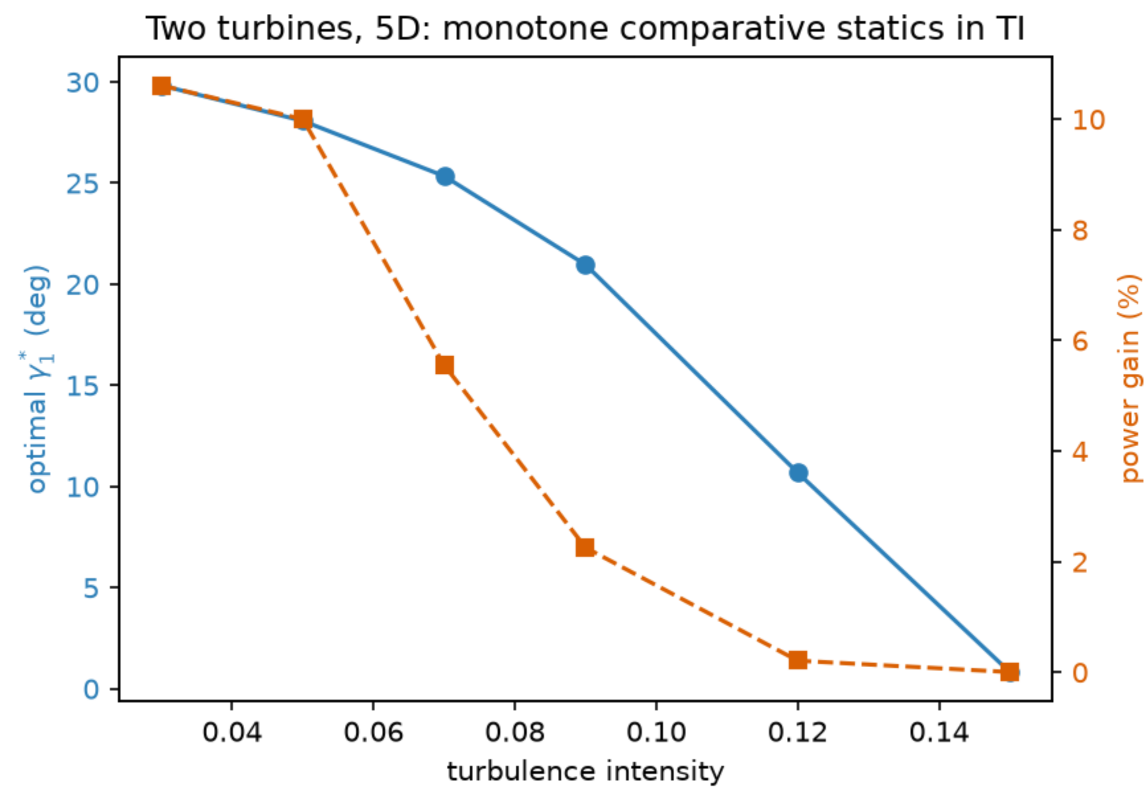

Figure 4: Comparative statics: the optimal yaw γ_1^* decreases monotonically from 29.8° to 0.8° as TI rises $0.03 \to 0.15$, exactly as the single-crossing argument predicts, with no convexity assumption.

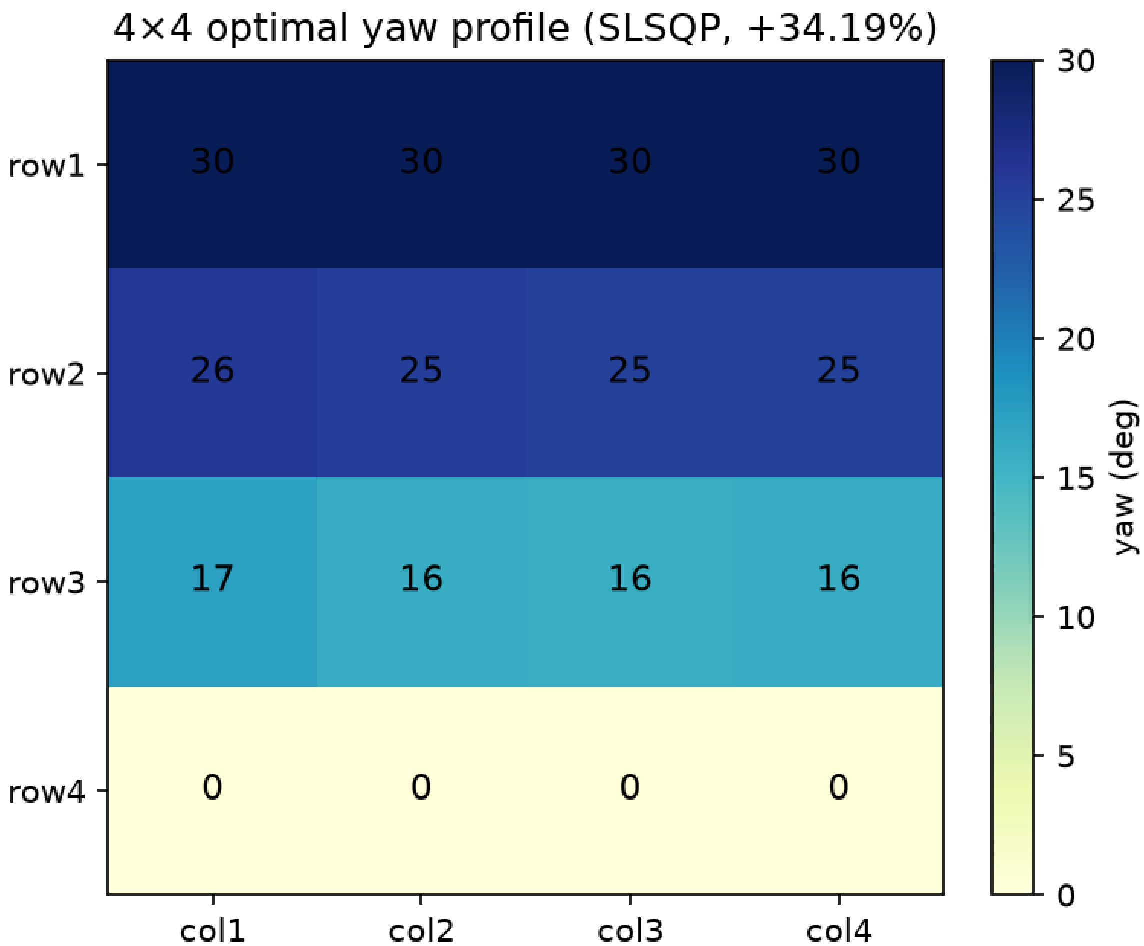

Figure 5: The 4×4 optimum is row-synchronous and monotonically decreasing downstream: $[30, 30, 30, 30]$, $[25.8, 25.2, 25.1, 25]$, $[17.2, 16, 15.7, 16]$, $[0, 0, 0, 0]$ degrees ($+34.19\,\%$).

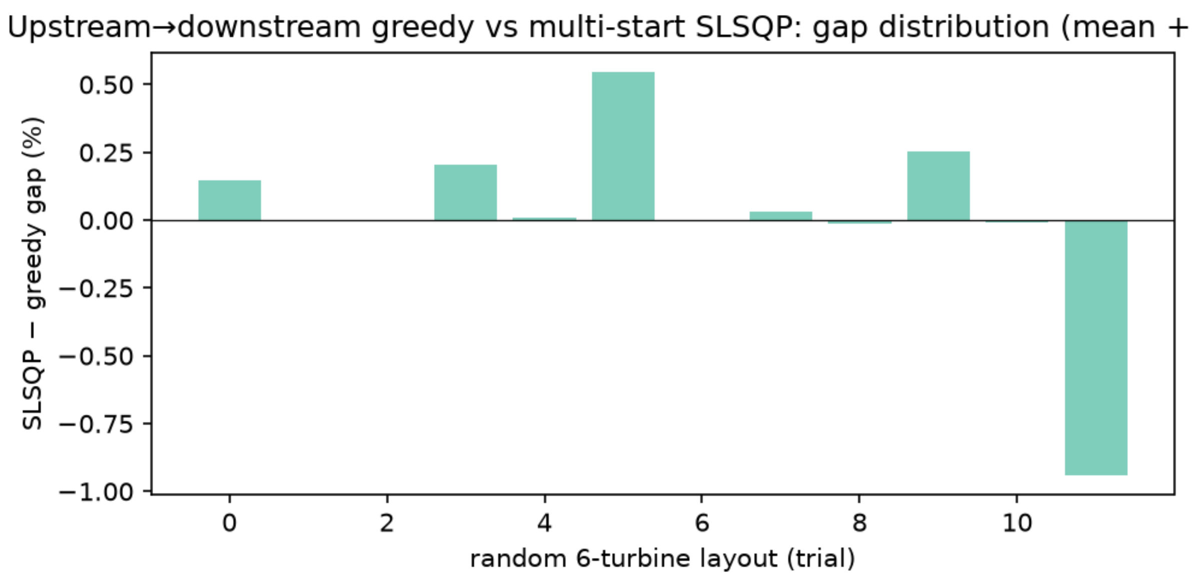

Figure 6: Greedy near-optimality and the bounded-interaction certificate: measured gaps (mean 0.103 %, max 0.477 %) sit below the sampled-interaction bound on all 12 random layouts.

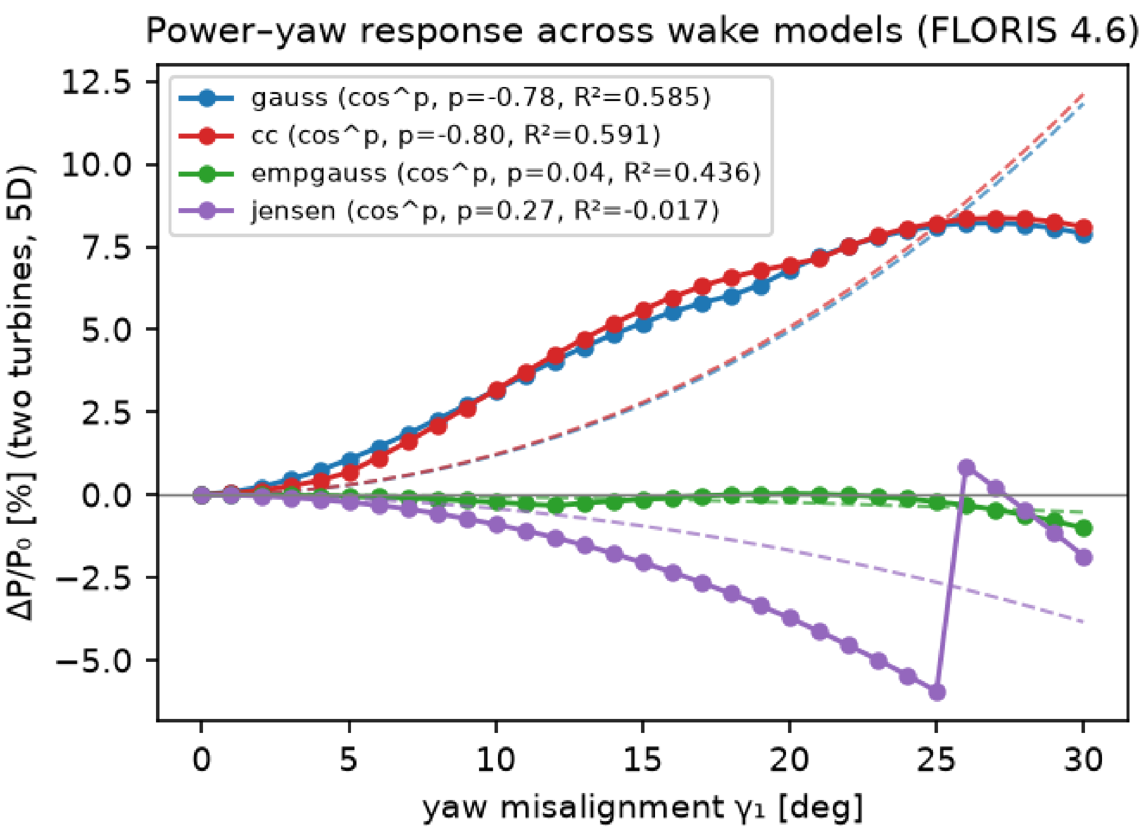

Figure 7: (a) Two-turbine farm response across four FLORIS model families; (b) the farm response decomposes into the exact $\cos^{1.88}$ self-cost term of the upstream turbine and the rising wake-recovery term of the downstream turbine.

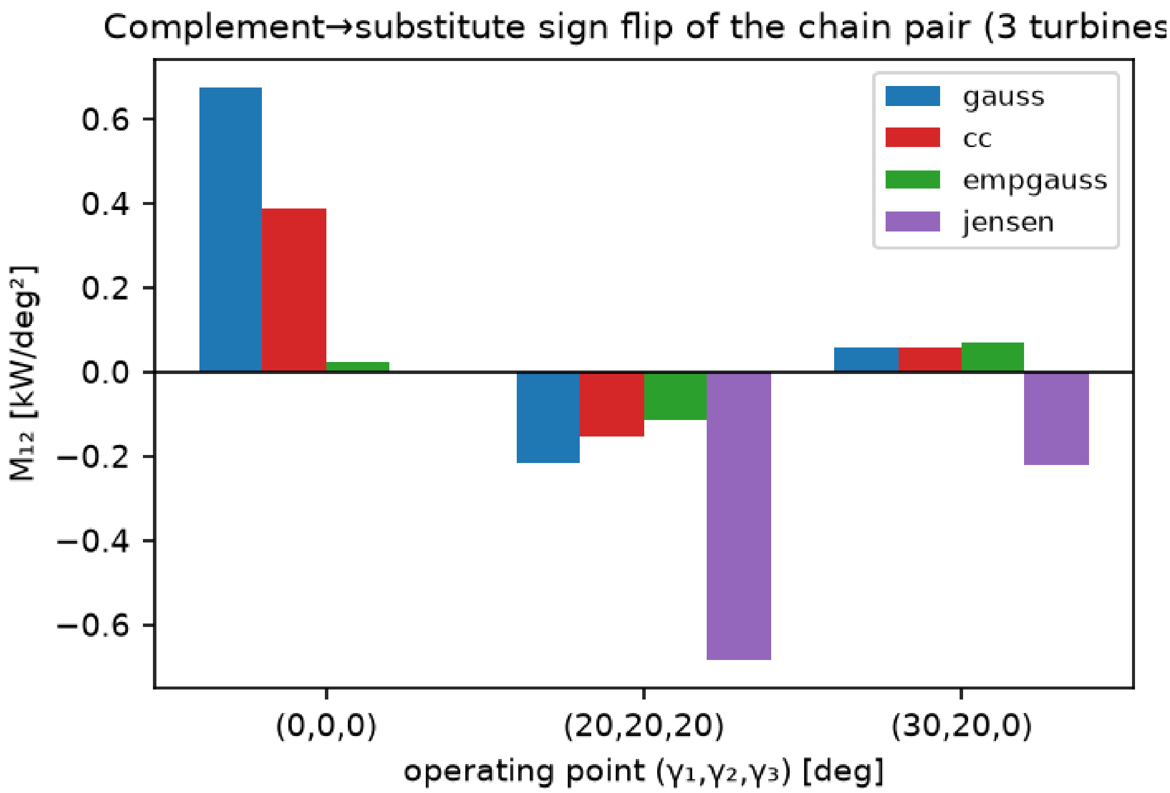

Figure 8: The chain-pair sign flip reproduces in the LES-calibrated *cc* and field-calibrated *empirical_gauss* models: $+0.674/+0.388/+0.022$ at the origin versus $-0.215/-0.154/-0.114\,\mathrm{kW\,deg^{-2}}$ at $(20°, 20°, 20°)$.

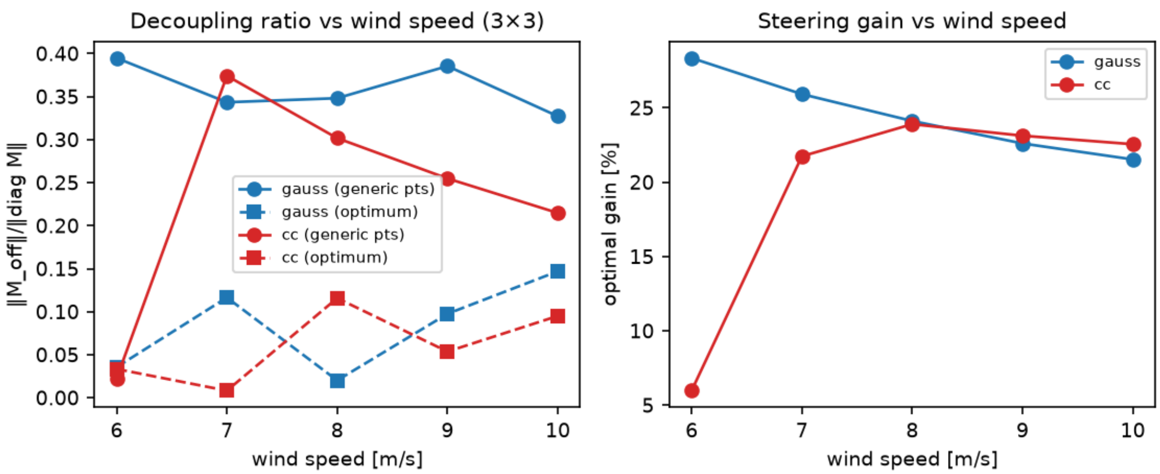

Figure 9: (a) Decoupling ratio at generic points versus optima across 6–10 $\mathrm{m\,s^{-1}}$ for gauss and cc; (b) steering gain versus wind speed. The decoupling law persists across the speed range.

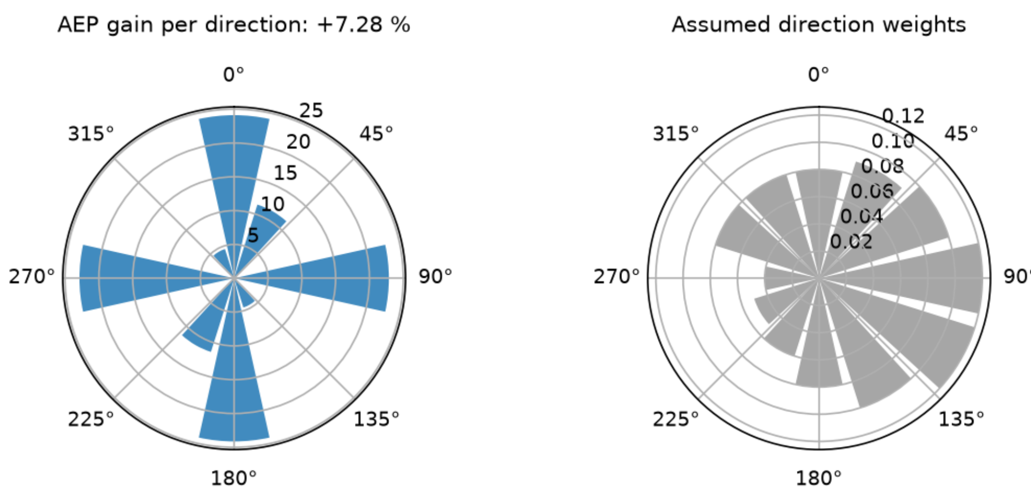

Figure 10: AEP gains per direction and the assumed direction weights: +7.28 % AEP over the 12-direction rose, the energy-side version of the decoupling law.

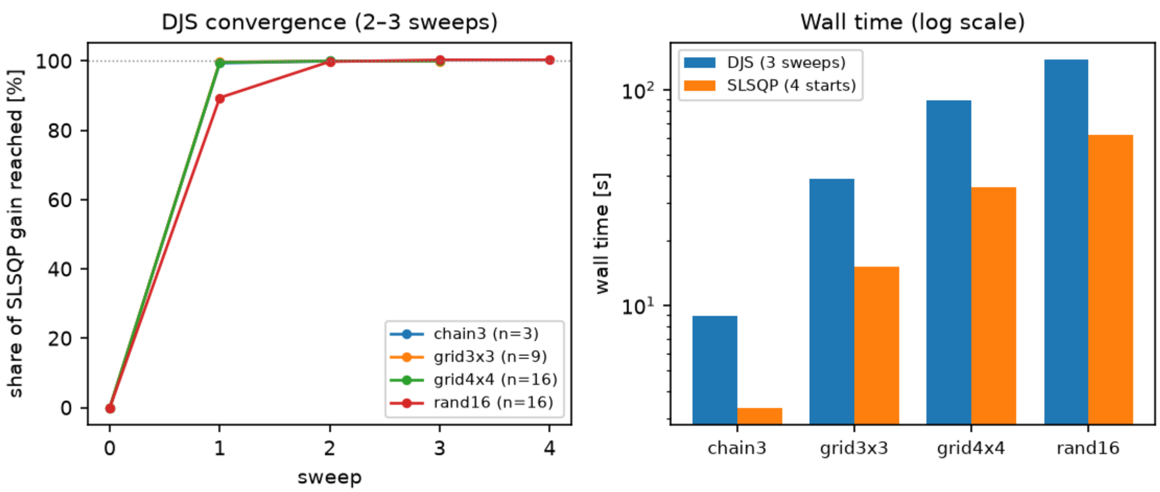

Figure 11: (a) Decoupled Jacobi sweeps reach $\geq 99\,\%$ of the SLSQP gain after one sweep and $\geq 99.8\,\%$ after two; (b) wall times at equal precision.

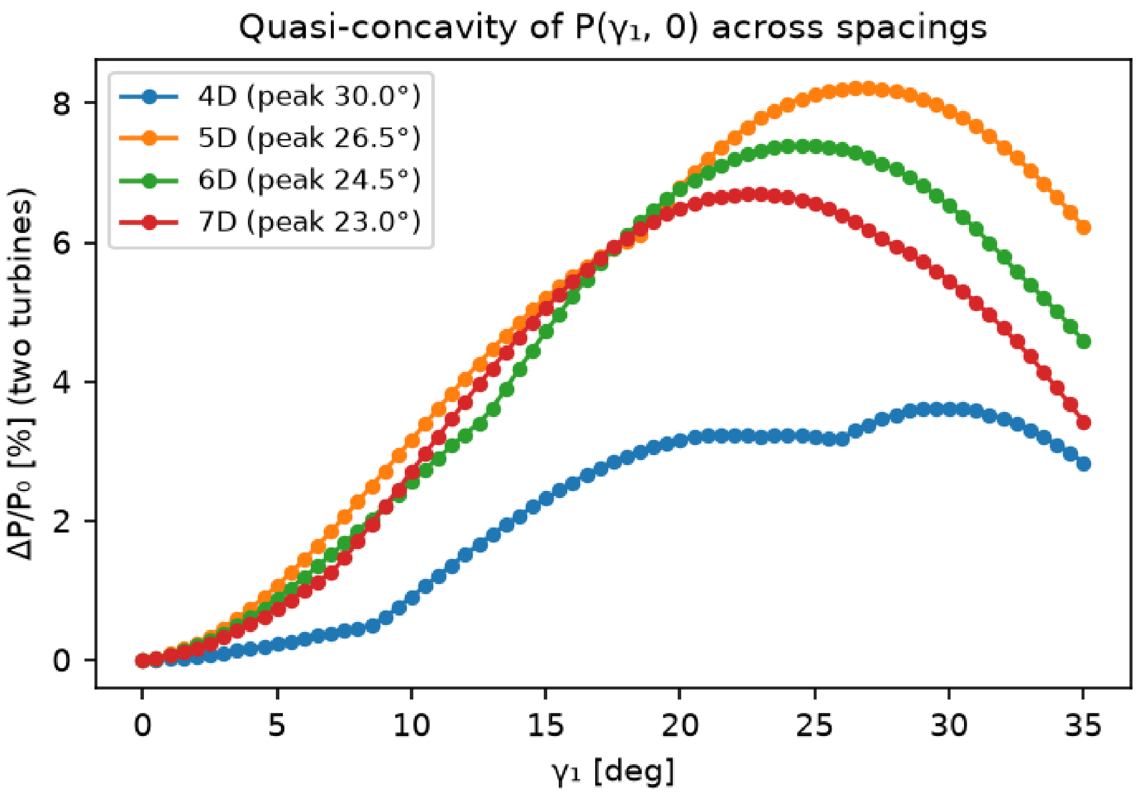

Figure 12: Two-turbine $P(\gamma_1, 0)$ is single-peaked for every spacing (0.5° grid; peaks $30^\circ/26.5^\circ/24.5^\circ/23^\circ$ at 4D/5D/6D/7D); the 4D tail beyond 25° lies in the excluded boundary zone (negative-velocity warnings).